bigger than 5 microns are likely to settle in the nasal passage whereas particles of about 10 micron enters the lungs easily and causes scaring or fibrosis of lung lining. They irritate the lungs and causes cancer and asthma. This disease is also called pneumoconiosis. Coal miners may suffer from black lung disease. Textile workers may suffer from white lung disease.

ii. Lead particulates affect children's brain, interferes maturation of RBCs and even cause cancer.

iii. Particulates in the atmosphere reduce visibility by scattering and absorption of sunlight. It is dangerous for aircraft and motor vehicles

iv. Particulates provide nuclei for cloud formation and increase fog and rain.

v. Particulates deposit on plant leaves and hinder the intake of CO_2 from the air and affect photosynthesis.

15.3.2.3. Techniques to reduce particulate pollutants

The particulates from air can be removed by using electrostatic precipitators, gravity settling chambers, and wet scrubbers or by cyclone collectors. These techniques are based on washing away or settling of the particulates.

15.3.3 Smog

Smog is a combination of smoke and fog which forms droplets that remain suspended in the air.



Fig 15.4 classical smog

Smog is a chemical mixture of gases that forms a brownish yellow haze over urban cities.Smog mainly consists of ground level ozone, oxides of nitrogen, volatile organic compounds, SO_2, acidic aerosols and gases, and particulate matter.

There are two types of smog. One is Classical smog caused by coal smoke and fog , second one is photo chemical smog caused by photo chemical oxidants.They are discussed below in detail.

(i) Classical smog or London smog

Classical smog was first observed in London in December 1952 and hence it is also known as London smog. It consists of coal smoke and fog.

It occurs in cool humid climate. This atmospheric smog found in many large cities. The chemical composition is the mixture of SO_2, SO_3 and humidity. It generally occurs in the morning and becomes worse when the sun rises.

This is mainly due to the induced oxidation of SO_2 to SO_3, which reacts with water yielding sulphuric acid aerosol.

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